

Bayesian log reg model with random effects

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Set working directory; adjust as necessary

Load brms for Bayesian regression with random effects

```
library(brms)
```

```
## Loading required package: Rcpp
## Loading 'brms' package (version 2.23.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
##
## Attaching package: 'brms'
##
## The following object is masked from 'package:stats':
##
##      ar
```

Load data

```
df = read.delim('data_Anna.csv', stringsAsFactors = TRUE)
df$sentence <- as.character(df$sentence)
df['outcome'] = df['construction'] == 'AA'
```

Functions for models

```
run_model <- function(frm)
{
  mod <- brm(
    frm,
    family = bernoulli(),
    data = df,
    chains = 4,
    cores = 4,
    iter = 4000,
    warmup = 2000,
    seed = 123,
    prior = c(
      prior(normal(0, 2.5), class = "b"),
      prior(normal(0, 5), class = "Intercept")
    ),
    control = list(adapt_delta = 0.95),
    save_pars = save_pars(all = TRUE)
  )
  return(mod)
}
```

```
run_model_re <- function(frm)
{
  mod <- brm(
    frm,
    family = bernoulli(),
    data = df,
    chains = 4,
    cores = 4,
    iter = 4000,
    warmup = 2000,
    seed = 123,
    prior = c(
      prior(normal(0, 2.5), class = "b"),
      prior(normal(0, 5), class = "Intercept"),
      prior(exponential(1), class = "sd")
    ),
    control = list(adapt_delta = 0.95),
    save_pars = save_pars(all = TRUE)
  )
  return(mod)
}
```

Fit models

```
# model without effects for author/work
frm <- formula("outcome ~ Position + information.Status +
               Topic.persistence + Discourse.Hierarchy + Agentivity")
mod_none <- run_model(frm)
```

```
## Compiling Stan program...
```

```
## Start sampling
```

```
# model with fixed effect for work
frm <- formula("outcome ~ Position + information.Status + Topic.persistence +
               Discourse.Hierarchy + Agentivity + work")
mod_fe_work <- run_model(frm)
```

```
## Compiling Stan program...
```

```
## Start sampling
```

```
# model with fixed effect for Author
frm <- formula("outcome ~ Position + information.Status + Topic.persistence +
               Discourse.Hierarchy + Agentivity + Author")
mod_fe_author <- run_model(frm)
```

```
## Compiling Stan program...
```

```
## Start sampling
```

```
# model with random effect for work
frm <- formula("outcome ~ Position + information.Status + Topic.persistence +
               Discourse.Hierarchy + Agentivity + (1|work)")
mod_re_work <- run_model_re(frm)
```

```
## Compiling Stan program...
```

```
## Start sampling
```

```
# model with random effect for Author
frm <- formula("outcome ~ Position + information.Status + Topic.persistence +
  Discourse.Hierarchy + Agentivity + (1|Author)")
mod_re_author <- run_model_re(frm)
```

```
## Compiling Stan program...
```

```
## Start sampling
```

```
## :
```

```
## Warning: There were 2 divergent transitions after warmup. See
```

```
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

```
## to find out why this is a problem and how to eliminate them.
```

```
## Warning: Examine the pairs() plot to diagnose sampling problems
```

```
# model with random effects for author
```

```
frm <- formula("outcome ~ Position + information.Status + Topic.persistence +
  Discourse.Hierarchy + Agentivity + (1|Author/work)")
```

```
mod_re_author_work <- run_model_re(frm)
```

```
## Compiling Stan program...
```

```
## Start sampling
```

```
## Warning: There were 1 divergent transitions after warmup. See
```

```
## https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

```
## to find out why this is a problem and how to eliminate them.
```

```
## Warning: Examine the pairs() plot to diagnose sampling problems
```

Model comparison

```
loo_none <- loo(mod_none, moment_match = TRUE)
```

```
loo_fe_work <- loo(mod_fe_work, moment_match = TRUE)
```

```
loo_fe_author <- loo(mod_fe_author, moment_match = TRUE)
```

```
loo_re_work <- loo(mod_re_work, moment_match = TRUE)
```

```
loo_re_author <- loo(mod_re_author, moment_match = TRUE)
```

```
loo_re_author_work <- loo(mod_re_author_work, moment_match = TRUE)
```

```
loo_compare(loo_none,
  loo_fe_work,
  loo_fe_author,
  loo_re_work,
  loo_re_author,
  loo_re_author_work)
```

```
##               elpd_diff se_diff
## mod_none         0.0      0.0
## mod_re_author    -0.3      0.2
## mod_re_work      -0.4      0.2
## mod_re_author_work -0.7      0.3
## mod_fe_author    -1.0      0.7
## mod_fe_work      -1.5      0.6
```

```
summary(mod_none)
```

```
## Family: bernoulli
```

```
## Links: mu = logit
```

```
## Formula: outcome ~ Position + information.Status + Topic.persistence + Discourse.Hierarchy + Agentiv
```

```
## Data: df (Number of observations: 160)
```

```
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Regression Coefficients:
##
##               Estimate Est.Error 1-95% CI u-95% CI Rhat
## Intercept      -2.46      1.08    -4.80    -0.51 1.00
## Position       -3.64      1.23    -6.11    -1.29 1.00
## information.Statusnew  4.01      1.07     2.00     6.24 1.00
## information.Statusrecap 3.16      1.36     0.55     5.95 1.00
## Topic.persistencediscontinuous 0.75      1.05    -1.28     2.79 1.00
## Topic.persistencesetting 3.23      1.23     0.94     5.75 1.00
## Discourse.HierarchyB   -2.88      1.03    -4.99    -0.89 1.00
## AgentivityD          3.76      1.00     1.87     5.83 1.00
##
##               Bulk_ESS Tail_ESS
## Intercept      8433      5637
## Position       6082      5931
## information.Statusnew  6105      5348
## information.Statusrecap 6502      5585
## Topic.persistencediscontinuous 7040      5861
## Topic.persistencesetting 5820      6155
## Discourse.HierarchyB   6104      5223
## AgentivityD          6535      5508
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(mod_none)
```



